

Raj Dineshbhai Adroja

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Education

San Jose State University, Master of Science (MS) in Software Engineering spec. in Data Science **Aug 2024 – May 2026**

• **Courses:** Software Systems Engineering, Machine Learning, Enterprise Software Platforms, Computer Networks

Vellore Institute of Technology, Vellore, India, B.Tech in Computer Science GPA: 3.83/4.00 **Jul 2018 – May 2022**

• **Courses:** Network Engineering, Artificial Intelligence, Operating Systems, Image Processing, Cyber Security, Digital Forensics

Technical Skills

Areas Worked: Data Engineering, Machine Learning, Network Engineering, Web Development

Programming: Java, SQL, Python, Bash, React, Node.js, HTML, CSS, JavaScript, Spark, C++ , C, Golang

Tools: HANA, JIRA, Snowflake, Docker, Kubernetes, GitHub, Cisco Packet Tracker, Postman, Hadoop, Kafka, HDFS, MATLAB, Hive

Professional Experience

Data Engineering Management & Governance Analyst, Accenture – Bangalore, India **Nov 2022 – Jun 2024**

- Spearheaded **SAP HANA** to **Snowflake** data migration which improved storage efficiency by 16% and query performance by 12%.
- Designed and developed end-to-end migration process for various object types, ensuring seamless transfer of complex objects, eliminating manual errors resulting in 30% efficiency gain.
- Automated validation process for 1000+ objects for reliable operations during & post migration and achieved 100% data accuracy.
- Utilized **Sqoop** to analyze and export data to relational databases, enabling visualization and report generation for the BI team.
- Developed data conversion scripts and modified interfaces with external systems affected by data warehouse integration.

Software Engineering Intern, Viviana – Vadodara, India **Apr 2021 – May 2022**

- Developed dynamic and responsive user interfaces using **React**, **Vite**, **JavaScript**, **HTML**, **CSS** improving user experience and website performance.
- Collaborated with front-end team to integrate interface with **RESTful APIs** to enhance functionality and user experience.
- Optimized web performance and accessibility by implementing best practices for **content delivery networks (CDNs)** resulting in faster load times by 21%.

Teaching Assistant, VIT – Vellore, Tamil Nadu **Sep 2019 – Dec 2019**

- Conducted group tutoring sessions with students for subjects like Data Structures and Algorithms, object-oriented programming.
- Helped professor with grading of assignments & exams. Provided continuous feedback to enhance student performance.

Academic Projects

Automatic colorization using auto-encoder

- Implemented deep learning technique to learn an efficient representation of images, utilizing 8-layer **neural network encoders** with **Keras** to extract mid-level image features such as edges and textures, using **NumPy** for numerical computations and **OpenCV** for image processing tasks.
- Achieved 4% reduction in both dimensionality and noise through advanced optimization techniques, enhancing the model's ability to focus on crucial image areas and improving the accuracy of the image colorization process, employing **Matplotlib** for visualizing results and performance metrics.

Sentimental Analysis of Airline Tweets

- Implemented and tested machine learning models such as Random Forest (**RF**), Support Vector Machine (**SVM**), Logistic Regression (**LR**), Term Frequency-Inverse Document Frequency (**TF-IDF**), K-Nearest Neighbors (**KNN**), and **AdaBoost** for sentiment analysis, achieving a 5% improvement in prediction accuracy over baseline models.
- Performed an in-depth analysis of key performance metrics, such as accuracy, precision, recall, and F1-score, to comprehensively assess effectiveness of each algorithm.
- Conducted comparative analysis using cross-validation and statistical tests, identifying Random Forest as top-performing algorithm with a 7% higher F1-score compared to average of other tested methods.

V-PROPEL portal

- Collaborated with university development team to design and build an examinations portal, implementing front-end components using **React**, **HTML**, **CSS**, and **JavaScript**.
- Developed key features enabling 5000+ students to upload assignments, take 200+ online tests annually, and submit project reports, resulting in a 6% increase in timely submissions.
- Integrated grading functionality for professors, reducing grading time by 11% and improving efficiency of reviewing assignments.